

Capacity of the Vector Gaussian Channel in the Small Amplitude Regime

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Problem of Interest

$$Y = X + Z \text{ where } Z \sim \mathcal{N}(0, I_n)$$

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Maximum Minimum Mean Square Error (MMSE)

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Reminder:

$$\text{mmse}(X|Y) = \mathbb{E} [\|X - \mathbb{E}[X|Y]\|^2]$$

Problems of Interest

$$Y = X + Z \text{ where } Z \sim \mathcal{N}(0, I_n)$$

Maximum Mutual Information

$$\max_{X: X \in \mathcal{B}_n(R)} I(X; Y)$$

- Maximizer = **Capacity Achieving Input Distributions (CAID)**
- The random variable distributed according to the CAID results in the **largest** transmission rate

Maximum Minimum Mean Square Error (MMSE)

$$\max_{X: X \in \mathcal{B}_n(R)} \text{mmse}(X|Y)$$

- Maximizer = **Least Favorable Prior Distribution (LFPD)**
- The random variable distributed according to the LFPD is the '**hardest**' estimate

Outline

- What is known about the two problems
- Our contributions
- Proof technique
- Outlook

What Do We Know?

Maximum Mutual Information

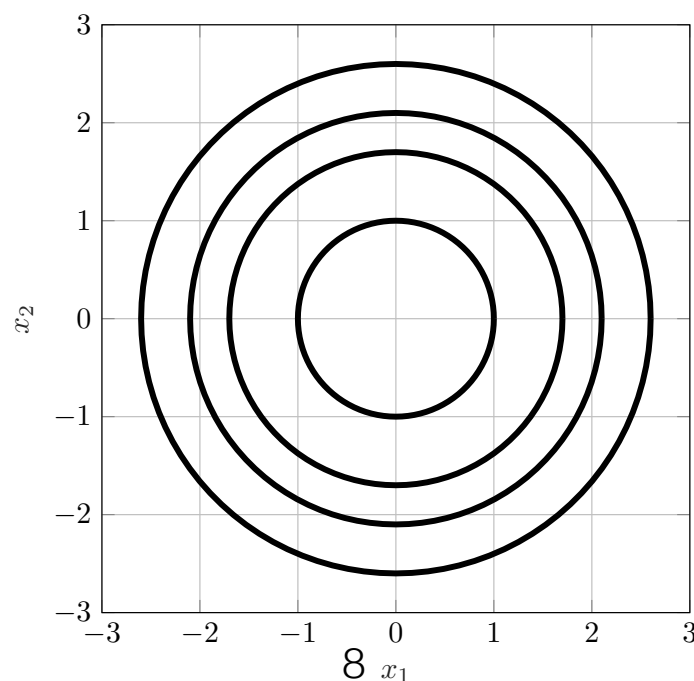
$$\max_{X: X \in \mathcal{B}_n(R)} I(X; Y)$$

- For $n=1$, the CAID is discrete with finitely many points. [Smith 1971]
- For $n \geq 2$, the CAID is supported on finitely many concentric spheres [Shamai, Bar-David 1995] [Rassouli, Clerckx 2016]

Maximum Minimum Mean Square Error (MMSE)

$$\max_{X: X \in \mathcal{B}_n(R)} \text{mmse}(X|Y)$$

- For $n=1$, the LFPD is discrete with finitely many points. [Ghosh 1969]
- For $n \geq 2$, the LFPD is supported on finitely many concentric spheres [Kempthorn 1986]



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- For $n=1$, the CAID is discrete with finitely many points. [Smith 1971]
- For $n \geq 2$, the CAID is supported on finitely many concentric spheres [Shamai, Bar-David 1995] [Rassouli, Clerckx 2016]
- For $n=1$, CAID is binary if and only if $R \leq 1.665$. [Sharma and Shamai 2010]
- For $n \geq 2$, the CAID is supported on a **single** sphere is open. Numerical results and discussion can be found in [Rassouli, Clerckx 2016]

Maximum Minimum Mean Square Error (MMSE)

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- For $n \geq 2$, the LFPD is supported on finitely many concentric spheres [Kempthorn 1986]
- For $n=1$, LFPD is binary if and only if $R \leq 1.057$ [Casella 1981]
- For $n \geq 2$, the LFPD is supported on a **single** sphere if and only if

$$R \leq \bar{R}_n^{\text{MMSE}} \approx 1.15\sqrt{n} \quad [\text{Berry 1990}]$$

Are LFPD's and CAID's Connected?

I-MMSE

$$I(X; Y) = \frac{1}{2} \int_0^1 \text{mmse}(X|Y_\gamma) d\gamma, \text{ where } Y_\gamma = \sqrt{\gamma}X + Z.$$

[Guo, Shamai, Verdu' 2005]

Maximum Mutual Information

$$\begin{aligned} \max_{X: X \in \mathcal{B}_n(R)} I(X; Y) &= \max_{X: X \in \mathcal{B}_n(R)} \frac{1}{2} \int_0^1 \text{mmse}(X|Y_\gamma) d\gamma \\ &\leq \frac{1}{2} \int_0^1 \max_{X: X \in \mathcal{B}_n(R)} \text{mmse}(X|Y_\gamma) d\gamma \end{aligned}$$

CAID = LFPD if and only if inequality is equality

Solution to the Max-MMSE

Maximum MMSE

Theorem: A single sphere is optimal if and only if

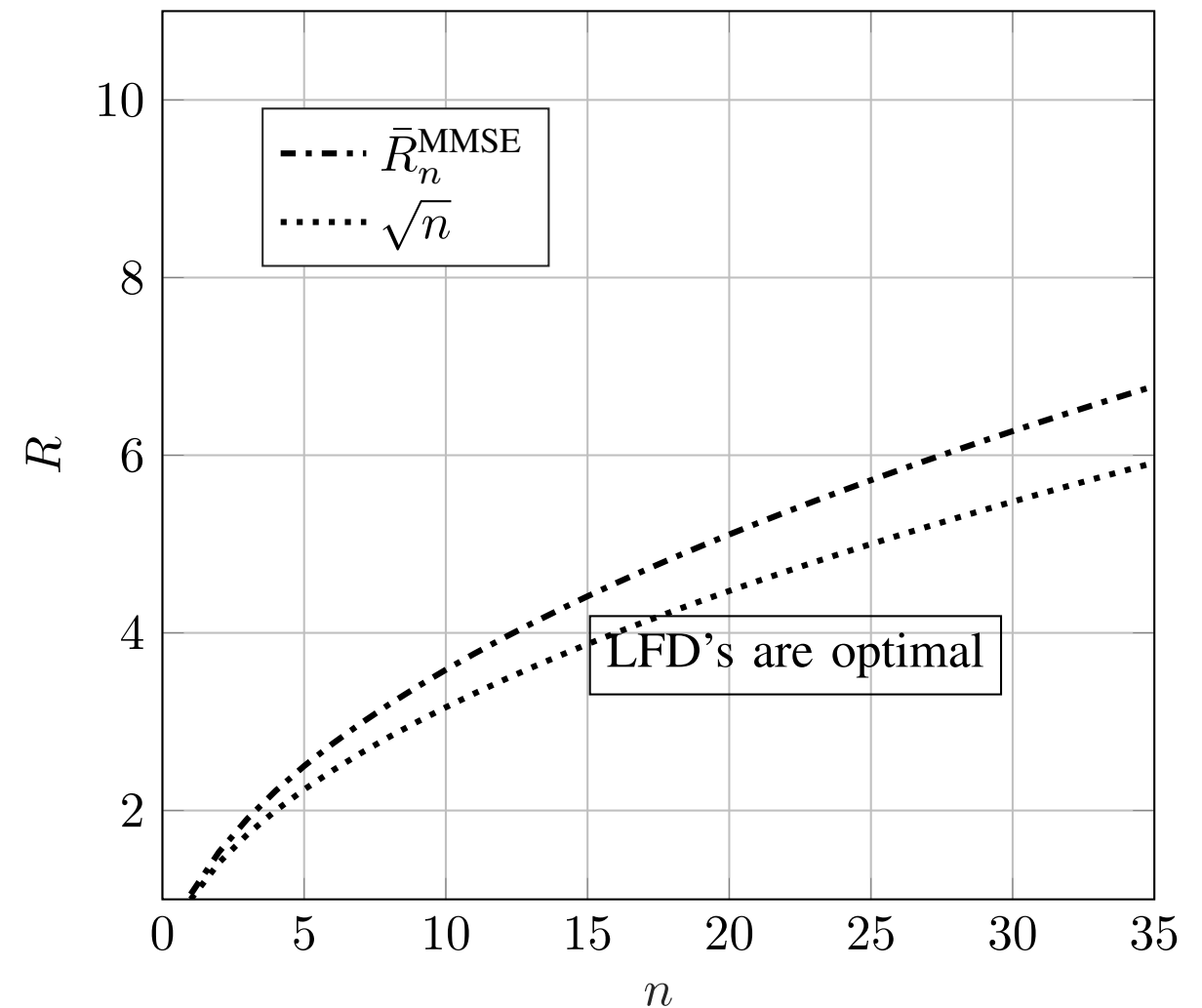
$$R \leq \bar{R}_n^{\text{MMSE}},$$

where $\bar{R}_n^{\text{MMSE}}$ is a zero of

$$f(\rho) = \mathbb{E} \left[h_{\frac{n}{2}}^2(\rho \|Z\|) \right] + \mathbb{E} \left[h_{\frac{n}{2}}^2(\rho \|x + Z\|) \right] - 1,$$

where $\|x\| = \rho$ and $h_v(t) = \frac{I_v(t)}{I_{v-1}(t)}$. **[Berry 1990]**

Maximum Mutual Information



Our Contribution

Maximum MMSE

Theorem: A single sphere is optimal if and only if

$$R \leq \bar{R}_n^{\text{MMSE}},$$

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where $\|x\| = \rho$ and $\mathbf{h}_v(t) = \frac{\mathbf{l}_v(t)}{\mathbf{l}_{v-1}(t)}$. **[Berry 1990]**

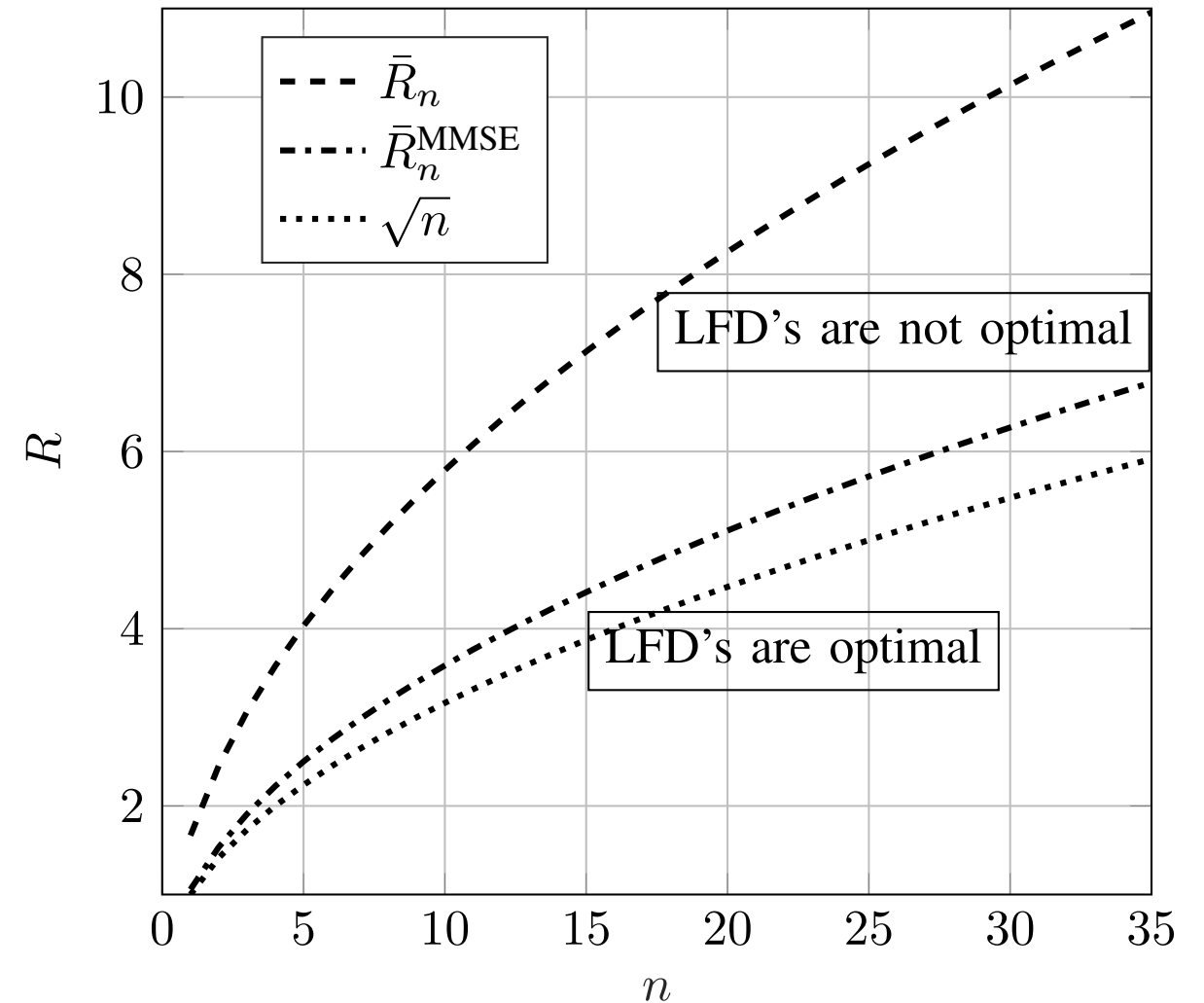
Maximum Mutual Information

Theorem: A single sphere is optimal if and only if

$$R \leq \bar{R}_n,$$

where $\bar{R}_n$ is a zero of

$$g(\rho) = \int_0^1 f(\sqrt{\gamma}\rho) d\gamma.$$



Proof Sketch

Step I: Characterize Conditions for the Optimality

P_X^* is a maximizer if and only if

$$\begin{aligned} i(x, P_X^*) &= I(P_X^*), \quad x \in \text{supp}(P_X^*), \\ i(x, P_X^*) &\leq I(P_X^*), \quad x \in \mathcal{B}_0(R). \end{aligned}$$

Where

$$i(x, P_X) = \mathbb{E} \left[\log \left(\frac{f_{Y|X}(Y|X)}{f_Y(Y)} \right) \mid X = x \right],$$

and

$$I(P_X) = \mathbb{E}[i(X, P_X)].$$

Step II: Find New Equivalent Conditions

P_X^* distributed on a single sphere is optimal if and only if for all $\|x\| = R$

$$i(0, P_X^*) \leq i(x, P_X^*).$$

Oscillation Theorem due to [Karlin 1957]

Proof Sketch

Step III: Use Pointwise I-MMSE

For every $R > 0$ and $\|x\| = R$

$$i(x, P_X^\star) = \frac{1}{2} \int_0^1 \mathbb{E} [\|X - \mathbb{E}[X \mid Y_\gamma]\|^2 \mid \|X\| = R] \, \mathrm{d}\gamma,$$

$$i(0, P_X^\star) = \frac{1}{2} \int_0^1 \mathbb{E} [\|X - \mathbb{E}[X \mid Y_\gamma]\|^2 \mid \|X\| = 0] \, \mathrm{d}\gamma,$$

Proof Sketch

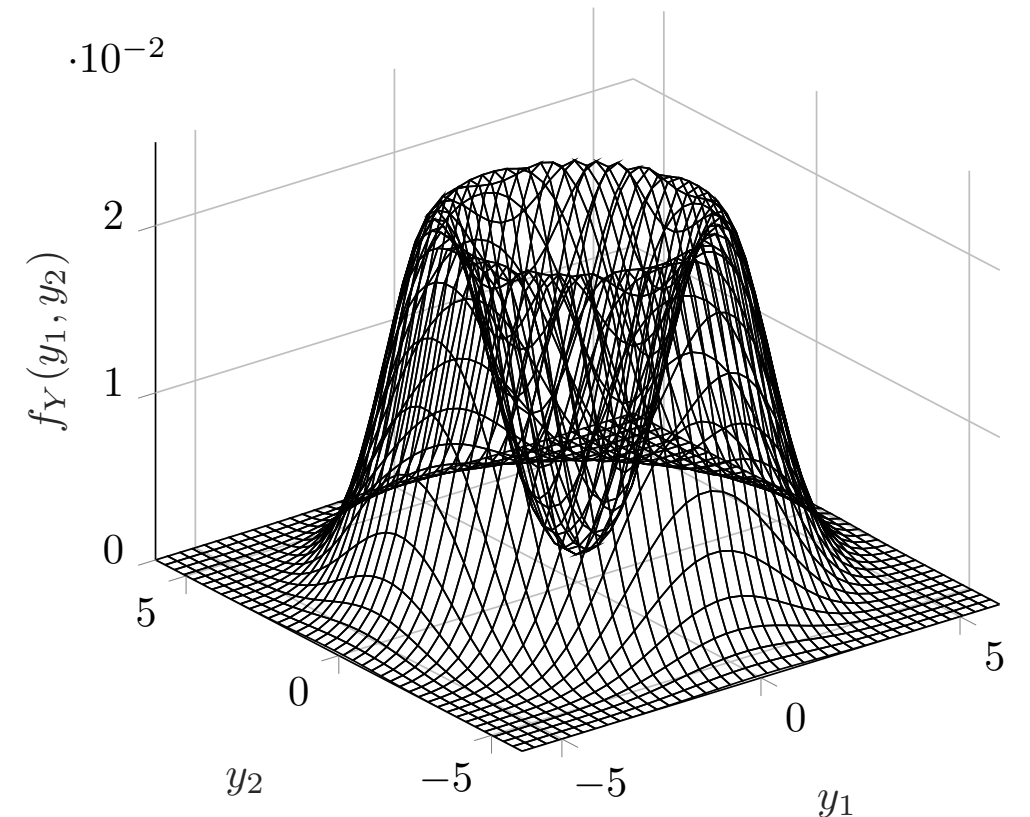
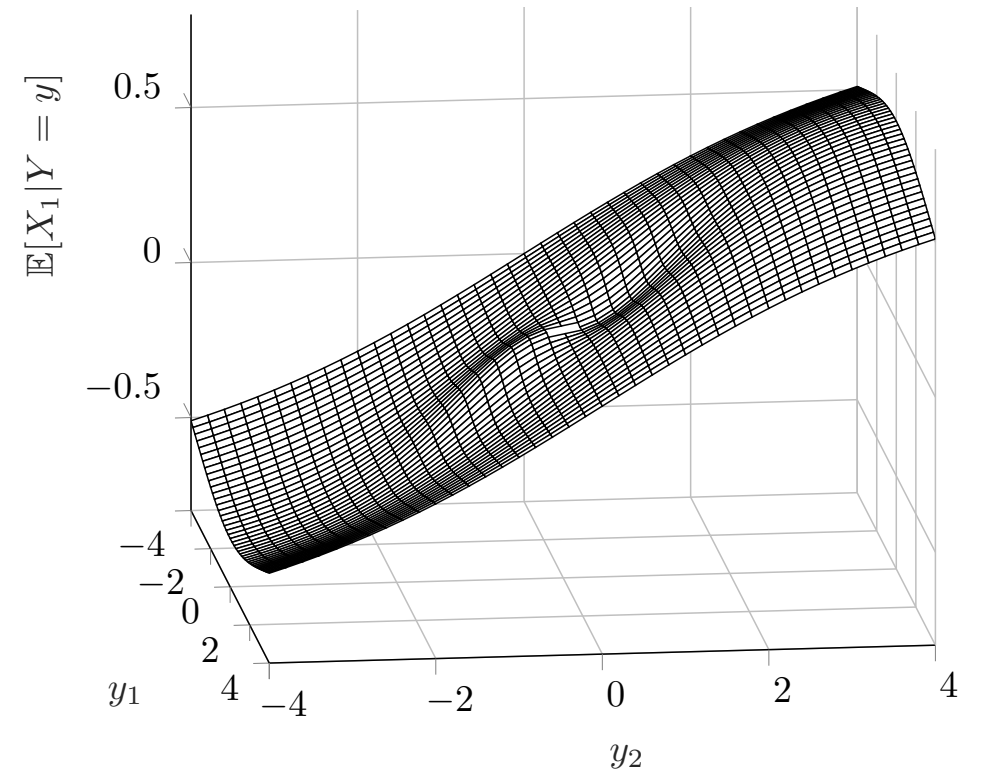
Step IV: Find Conditional Expectation for Spherical Shells

$$\mathbb{E}[X \mid Y = y] = \frac{Ry}{\|y\|} h_{\frac{n}{2}}(\|y\|R).$$

$$\mathbb{E}[X \mid Y = y] = y + \frac{\nabla_y f_Y(y)}{f_Y(y)}$$

[Esposito 1968]

$$f_Y(y) = \frac{\Gamma\left(\frac{n}{2}\right) e^{-\frac{R^2 + \|y\|^2}{2}}}{2\pi^{\frac{n}{2}}} \frac{I_{\frac{n}{2}-1}(\|y\|R)}{(\|y\|R)^{\frac{n}{2}-1}}.$$



Proof Sketch

Step V: Combine Step II, Step III and Step IV

$$\begin{aligned} 0 &\geq i(0, P_X^\star) - i(x, P_X^\star) \\ &= \int_0^1 \mathbb{E} [\|X - \mathbb{E}[X \mid Y_\gamma]\|^2 \mid \|X\| = 0] \\ &\quad - \mathbb{E} [\|X - \mathbb{E}[X_R \mid Y_\gamma]\|^2 \mid \|X\| = R] \, \mathrm{d}\gamma \\ &= \int_0^1 R^2 \mathbb{E} \left[\mathbf{h}_{\frac{n}{2}}^2 (\sqrt{\gamma} R \|Z\|) \right] - R^2 - R^2 \mathbb{E} \left[\mathbf{h}_{\frac{n}{2}}^2 (\sqrt{\gamma} R \|\sqrt{\gamma} x + Z\|) \right] \, \mathrm{d}\gamma \\ &= R^2 \int_0^1 f(\sqrt{\gamma} R) \, \mathrm{d}\gamma \end{aligned}$$

Proof Sketch

Step I: Characterize Conditions for the Optimality

Step II: Find New Equivalent Conditions. Use Oscillation Theorem

Step III: Use Pointwise I-MMSE

Step IV: Find Conditional Expectation for Spherical Shells

Step V: Combine Step II, Step III and Step IV

Outlook

- Found sufficient and necessary conditions for the optimality of a single sphere for every n
- The proof uses pointwise I-MMSE and Oscillation Theorem
- Oscillation Theorem can be used to show a bound on the number of spheres (Please come to our poster!)

Thank You!

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